

1. What is the total amount of Physical Memory (KB) available on your system (In use + Available)?

The total amount of Physical Memory for my system is 31,700,000 KB

2. Based on changes to the amount of Available memory, what is the apparent footprint (i.e., the full memory demand) of MS Edge with several web pages open?

Before I started MS Edge and I opened 10+ new pages my available memory was 16.9 GB. After completing that task, my available memory decreased to 13.9 GB

3. One of the measured components of available memory on Windows is the Standby list. What memory management mechanism described in your textbook does the Windows Standby list implement?

It uses a page-buffering algorithm that uses a pool of frames and a page is read into the pool before the victim frame is written out. It allows processes to restart as soon as possible. After the victim frame is written out, it gets added to a free-frame pool.

4.

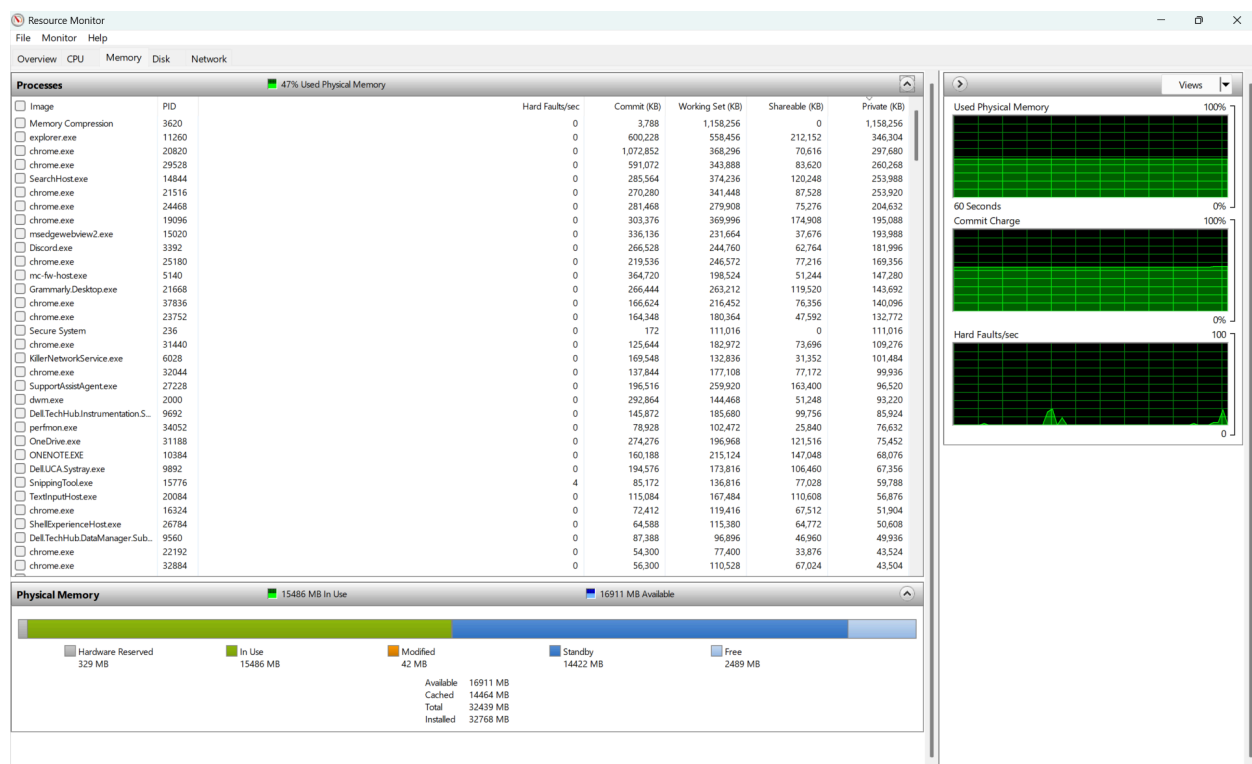


Figure 1. In use before MS edge tabs open

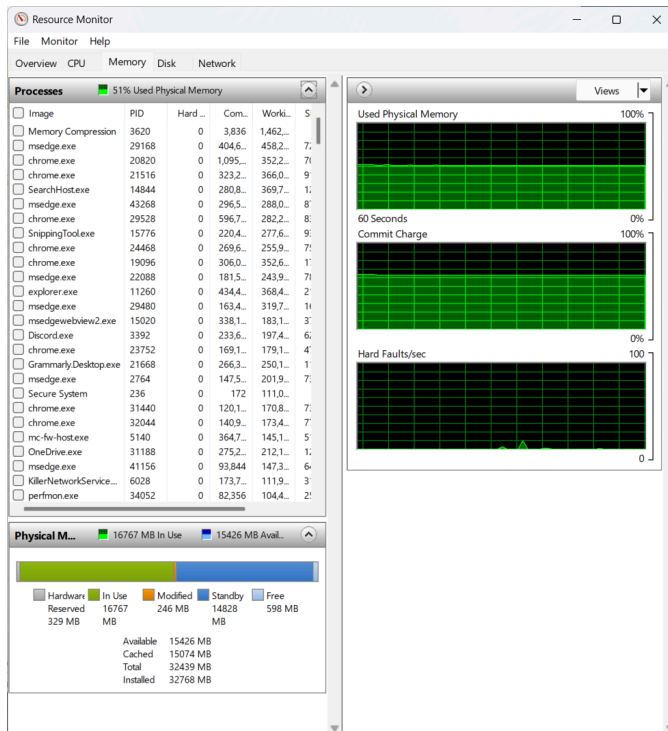


Figure 2. In use memory with 1 Edge tab open

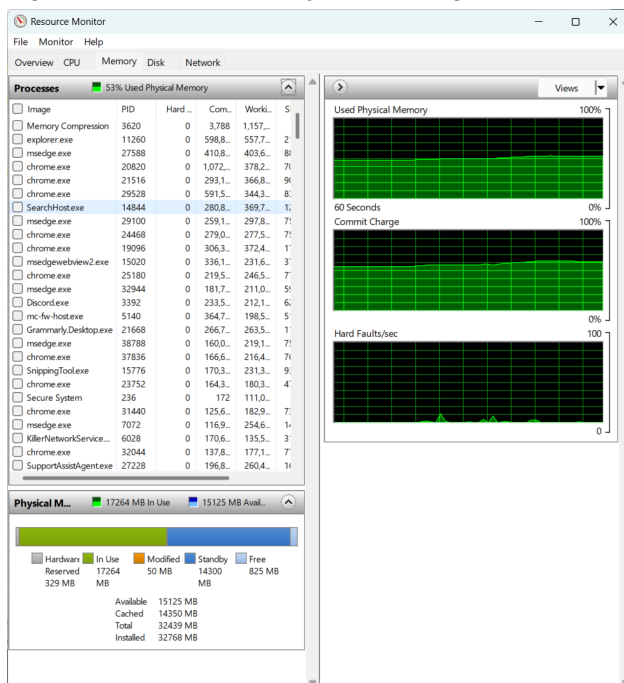


Figure 3. In use memory with 2 Edge tabs open

5. What are the amounts of Total Physical Memory and Total Virtual Memory available on your system?

Total physical memory for my system is 37.1 GB

Total Virtual Memory is 33.7 GB.

Virtual memory is a system used by operating systems to simulate a larger amount of memory, which allows a process to access memory that only exists as an extension as a part of the real main memory.

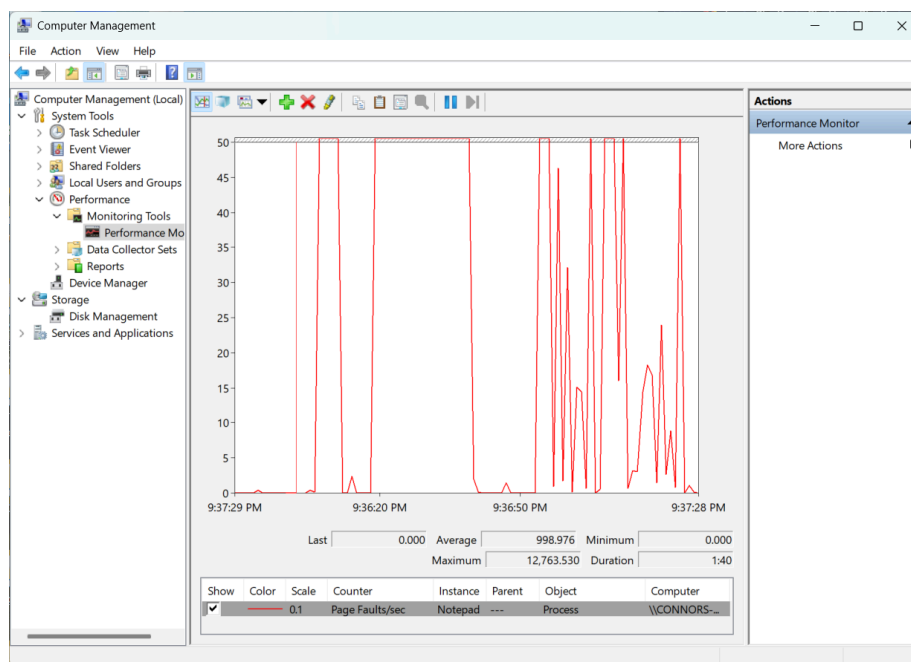
6. Under Virtual Memory, observe the size of the paging file

The paging file size is 2048MB which would be the difference between total physical memory and total virtual memory as observed in question 5.

7. How much virtual memory is File Explorer (explorer.exe) using?

Commit(Kb) is the amount of virtual memory reserved for the processes

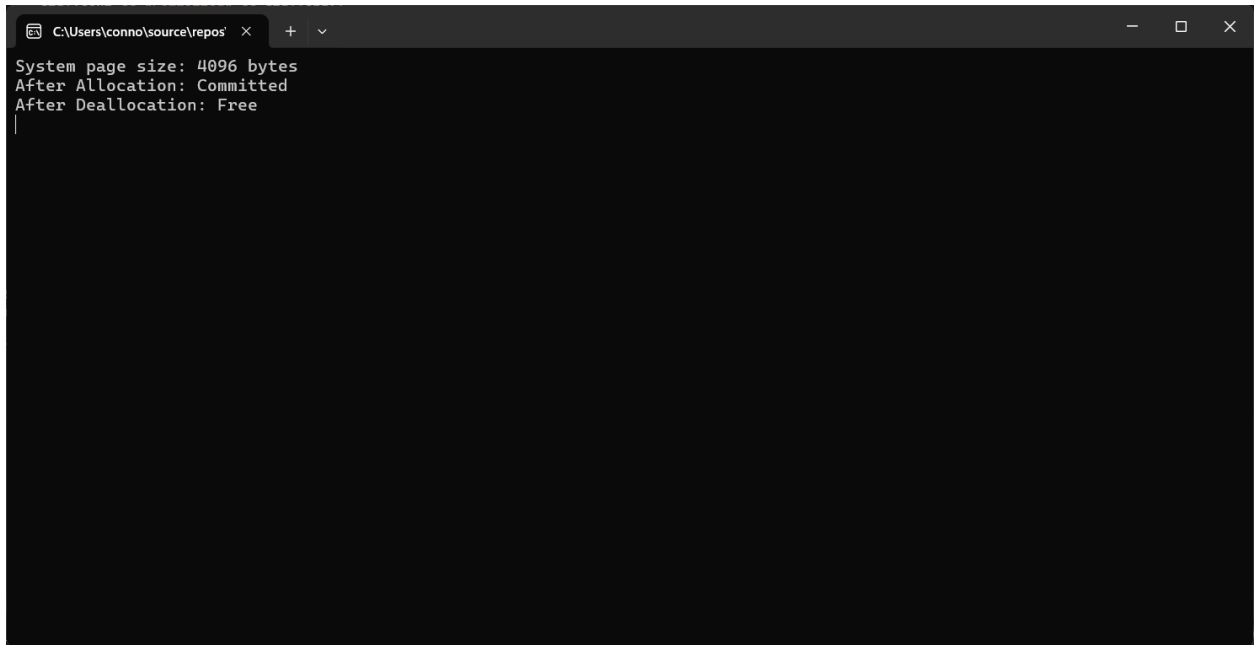
It is using 440,000 Kb of virtual memory.



8.

This is showing the windows operating system is using virtual memory to help the process Notepad. Pages and faults are used to help handle the dynamic memory requirement. The spikes in the graph show when the use is typing, selecting, or changing the fonts, these actions increase the memory demand as the process is changing.

Programming Assignment



```
C:\Users\conno\source\repos > System page size: 4096 bytes
After Allocation: Committed
After Deallocation: Free
```

The image shows a Windows command prompt window with a dark background. The title bar at the top indicates the current directory is C:\Users\conno\source\repos. The command prompt displays three lines of text: 'System page size: 4096 bytes', 'After Allocation: Committed', and 'After Deallocation: Free'. A vertical cursor is visible on the line following the last line of text.